

AMRITA VISHWA VIDYAPEETHAM CHENNAI
CAMPUS

Amrita School of Computing

Coding Practice

Activity Sheets

(For the Batch 2025-2029)

Name:

Roll No.:

Department:

Instructions

The **Mock Test syllabus** for CSE CT | CYS | AIE | RAI program is enclosed in the attached document for your reference.

Materials for the Mock Test [Must-DO Coding Questions] is provided in a link.

It is mandatory to upload the **completed Activity sheets** before the deadline in the MS-Form link shared by your class Advisor.

Activity Sheet: 01 - Coding Questions from Arrays – Part 1 [Easy + Medium Questions]

Each Activity Sheets contains total of **20 Questions**.

Platform: Any Online Compiler of your choice – **GDB Compiler**

Language: C | C++ | JAVA | PYTHON

The Activity Sheet will have the coding Questions and the Test Case. Once executed in Online Compiler, take **screen shot of the code and the output** and paste it in the space provided in the activity sheet.

The Final Activity Sheet should be in **PDF Format** and the file name should be your **roll number**.

Course Instructors: CARE TEAM

Dr.R.Annamalai	Assistant Professor (Sl.Grade) CSE-AIE
Dr.J.Umamageswaran	Assistant Professor (Sr.Grade) CSE-CT
Ms.D. Sasikala	Assistant Professor CSE-AIE
Mr.NirmalRaj	Assistant Professor CSE-CYS

1. You are given an integer array `arr[]`. You need to find the maximum sum of a subarray (containing at least one element) in the array `arr[]`.

Note : A subarray is a continuous part of an array.

Examples:

Input: `arr[] = [2, 3, -8, 7, -1, 2, 3]`

Output: 11

Explanation: The subarray `[7, -1, 2, 3]` has the largest sum 11.

Input: `arr[] = [-2, -4]`

Output: -2

Explanation: The subarray `[-2]` has the largest sum -2.

Input: `arr[] = [5, 4, 1, 7, 8]`

Output: 25

Explanation: The subarray `[5, 4, 1, 7, 8]` has the largest sum 25.

Constraints:

$1 \leq \text{arr.size()} \leq 10^5$

$-10^4 \leq \text{arr}[i] \leq 10^4$

Code:

Output:

2. You are given an array `arr[]` of size $n - 1$ that contains distinct integers in the range from 1 to n (inclusive). This array represents a permutation of the integers from 1 to n with one element missing. Your task is to identify and return the missing element.

Examples:

Input: `arr[] = [1, 2, 3, 5]`

Output: 4

Explanation: All the numbers from 1 to 5 are present except 4.

Input: `arr[] = [8, 2, 4, 5, 3, 7, 1]`

Output: 6

Explanation: All the numbers from 1 to 8 are present except 6.

Input: `arr[] = [1]`

Output: 2

Explanation: Only 1 is present so the missing element is 2.

Constraints:

$1 \leq \text{arr.size()} \leq 10^6$

$1 \leq \text{arr}[i] \leq \text{arr.size()} + 1$

Code:

Output

3. You are given a binary number as a string of characters ('0' and '1'). Your task is to determine whether this binary number is divisible by 3.

Note: Try to accomplish this using a single traversal of the input binary string.

Examples:

Input: `s = "100"`

Output: false

Explanation: "100"'s decimal equivalent is 4, which is not divisible by 3.

Input: `s = "0011"`

Output: true

Explanation: "0011" is 3, which is divisible by 3.

Input: `s = "110"`

Output: true

Explanation: The decimal equivalent of "110" is 6, which is divisible by 3.

Constraints:

$1 \leq s.size() \leq 10^6$

The string `s` contains only '0' and '1'.

Code:

Output

4. Given an array `arr[]` of integers and another integer `target`. Determine if there exist two distinct indices such that the sum of their elements is equal to the target.

Examples:

Input: `arr[] = [0, -1, 2, -3, 1]`, `target = -2`

Output: true

Explanation: `arr[3] + arr[4] = -3 + 1 = -2`

Input: `arr[] = [1, -2, 1, 0, 5]`, `target = 0`

Output: false

Explanation: None of the pair makes a sum of 0

Input: `arr[] = [11]`, `target = 11`

Output: false

Explanation: No pair is possible as only one element is present in `arr[]`

Constraints:

$1 \leq \text{arr.size} \leq 10^5$

$-10^5 \leq \text{arr}[i] \leq 10^5$

$-2 \cdot 10^5 \leq \text{target} \leq 2 \cdot 10^5$

Code:

Output

5. Given an integer array `arr`, return all the unique pairs `[arr[i], arr[j]]` such that $i \neq j$ and $arr[i] + arr[j] == 0$.

Note: The pairs must be returned in sorted order, the solution array should also be sorted, and the answer must not contain any duplicate pairs.

Examples:

Input: `arr = [-1, 0, 1, 2, -1, -4]`

Output: `[[-1, 1]]`

Explanation: $arr[0] + arr[2] = (-1) + 1 = 0$.

$arr[2] + arr[4] = 1 + (-1) = 0$.

The distinct pair are `[-1, 1]`.

Input: `arr = [6, 1, 8, 0, 4, -9, -1, -10, -6, -5]`

Output: `[[-6, 6], [-1, 1]]`

Explanation: The distinct pairs are `[-1, 1]` and `[-6, 6]`.

Constraints:

$3 \leq arr.size \leq 10^5$

$-10^5 \leq arr[i] \leq 10^5$

Code:

Output

6. Given a matrix Arr of size N x M. You are given position of submatrix as X1, Y1 and X2, Y2 inside the matrix. Find the sum of all elements inside that submatrix. Here X1, Y1, X2, Y2 are 1-based.

Example 1:

Input:

N = 5 , M = 6

Arr[][] = {{1, 2, 3, 4, 5, 6},
 {7, 8, 9, 10, 11, 12},
 {13, 14, 15, 16, 17, 18},
 {19, 20, 21, 22, 23, 24},
 {25, 26, 27, 28, 29, 30}}

X1=3, Y1=4, X2=4, Y2=5

Output: 78

Explanation: Sum from cell starting at position (3, 4) (1-based indexing) and ending at (4, 5) is 78 (16+17+22+23).

Example 2:

Input:

N = 3, M = 3

Arr[][] = {{9, 8, 7},{4, 2, 1},{6, 5, 3}}
X1=1, Y1=2, X2=3, Y2=3

Output: 26

Explanation: Sum from cell starting at position (1, 2) (1-based indexing) and ending at (3, 3) is 26.

Your Task:

You don't need to read input or print anything. Your task is to complete the function subMatrixSum() which takes the array of booleans arr[], n, m, x1, y1, x2 and y2 as parameters and returns an integer denoting the answer.

Expected Time Complexity: $O(N \cdot M)$

Expected Auxiliary Space: $O(1)$

Constraints:

$1 \leq N, M \leq 10^3$

$1 \leq \text{Arr}[N][M] \leq 10^6$

$1 \leq X1, X2 \leq N$

$1 \leq Y1, Y2 \leq M$

Code:

Output

7. Given an array `arr[]` . Rotate the array to the left (counter-clockwise direction) by `d` steps, where `d` is a positive integer. Do the mentioned change in the array in place.

Note: Consider the array as circular.

Examples :

Input: `arr[] = [1, 2, 3, 4, 5]`, `d = 2`

Output: `[3, 4, 5, 1, 2]`

Explanation: when rotated by 2 elements, it becomes `[3, 4, 5, 1, 2]`.

Input: `arr[] = [2, 4, 6, 8, 10, 12, 14, 16, 18, 20]`, `d = 3`

Output: `[8, 10, 12, 14, 16, 18, 20, 2, 4, 6]`

Explanation: when rotated by 3 elements, it becomes `[8, 10, 12, 14, 16, 18, 20, 2, 4, 6]`.

Input: `arr[] = [7, 3, 9, 1]`, `d = 9`

Output: `[3, 9, 1, 7]`

Explanation: when we rotate 9 times, we'll get `[3, 9, 1, 7]` as resultant array.

Constraints:

$1 \leq \text{arr.size}(), d \leq 10^5$

$0 \leq \text{arr}[i] \leq 10^5$

Code:

Output

8. Given the root of a Binary Tree, your task is to return its Level Order Traversal.

Note: A level order traversal is a breadth-first search (BFS) of the tree. It visits nodes level by level, starting from the root, and processes all nodes from left to right within each level before moving to the next.

Examples:

Input: root = [1, 2, N, N, 3, N, N]

Output: [1, 2, 3]

Explanation: We start with the root node 1, so the first level of the traversal is [1]. Then we move to its children 2 and 3, which form the next level, giving the final output [1, 2, 3].

Input: root = [10, 20, 30, N, N, 40, 50, N, N, N, N]

Output: [10, 20, 30, 40, 50]

Explanation: We begin with the root node 10, which forms the first level as [10]. Its children 20 and 30 make up the second level, and their children 40 and 50 form the third level, resulting in [10, 20, 30, 40, 50].

Constraints:

$1 \leq \text{number of nodes} \leq 3 \cdot 10^4$

$0 \leq \text{node} \rightarrow \text{data} \leq 10^9$

Code:

Output

9. Given an array `arr[]` containing $2*n + 2$ positive numbers, out of which $2*n$ numbers exist in pairs whereas only two number occur exactly once and are distinct. Find the other two numbers. Return the answer in increasing order.

Examples:

Input: `arr[] = [1, 2, 3, 2, 1, 4]`

Output: `[3, 4]`

Explanation: 3 and 4 occur exactly once.

Input: `arr[] = [2, 1, 3, 2]`

Output: `[1, 3]`

Explanation: 1 and 3 occur exactly once.

Input: `arr[] = [2, 1, 3, 3]`

Output: `[1, 2]`

Explanation: 1 and 2 occur exactly once.

Constraints:

$2 \leq \text{arr.size()} \leq 10^6$

$1 \leq \text{arr}[i] \leq 5 * 10^6$

`arr.size()` is even

Code:

Output

10. Given an array `arr` of positive integers. You have to return the maximum of $j - i$ such that $arr[i] < arr[j]$ and $i < j$.

Examples:

Input: `arr[] = [1, 10]`

Output: 1

Explanation: $arr[0] < arr[1]$ so $(j-i)$ is $1-0 = 1$.

Input: `arr[] = [5, 4, 3]`

Output: 0

Explanation: There is no pair that satisfies the given condition.

Input: `arr[] = [34, 8, 10, 3, 2, 80, 30, 33, 1]`

Output: 6

Explanation: In the given array $arr[1] < arr[7]$ satisfying the required condition ($arr[i] < arr[j]$) thus giving the maximum difference of $j - i$ which is $6(7-1)$.

Constraints:

$1 \leq arr.size \leq 10^5$

$0 \leq arr[i] \leq 10^9$

Code:

Output

11. Given an array `arr[]` containing only non-negative integers, your task is to find a continuous subarray (a contiguous sequence of elements) whose sum equals a specified value `target`. You need to return the 1-based indices of the leftmost and rightmost elements of this subarray. You need to find the first subarray whose sum is equal to the target.

Note: If no such array is possible then, return `[-1]`.

Examples:

Input: `arr[] = [1, 2, 3, 7, 5]`, `target = 12`

Output: `[2, 4]`

Explanation: The sum of elements from 2nd to 4th position is 12.

Input: `arr[] = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]`, `target = 15`

Output: `[1, 5]`

Explanation: The sum of elements from 1st to 5th position is 15.

Input: `arr[] = [5, 3, 4]`, `target = 2`

Output: `[-1]`

Explanation: There is no subarray with sum 2.

Constraints:

$1 \leq \text{arr.size()} \leq 10^6$

$0 \leq \text{arr}[i] \leq 10^3$

$0 \leq \text{target} \leq 10^9$

Code:

Output

12. Given an array of strings `arr[]`. Return the longest common prefix among each and every strings present in the array. If there's no prefix common in all the strings, return `""`.

Examples :

Input: `arr[] = ["geeksforgeeks", "geeks", "geek", "geezer"]`

Output: `"gee"`

Explanation: `"gee"` is the longest common prefix in all the given strings.

Input: `arr[] = ["hello", "world"]`

Output: `""`

Explanation: There's no common prefix in the given strings.

Constraints:

$1 \leq |arr| \leq 10^3$

$1 \leq |arr[i]| \leq 10^3$

Code:

Output

13. You are given a sorted array `arr[]` containing positive integers. Your task is to remove all duplicate elements from this array such that each element appears only once. Return an array containing these distinct elements in the same order as they appeared.

Examples :

Input: `arr[] = [2, 2, 2, 2, 2]`

Output: `[2]`

Explanation: After removing all the duplicates only one instance of 2 will remain i.e. `[2]` so modified array will contain 2 at first position and you should return array containing `[2]` after modifying the array.

Input: `arr[] = [1, 2, 4]`

Output: `[1, 2, 4]`

Explanation: As the array does not contain any duplicates so you should return `[1, 2, 4]`.

Constraints:

$1 \leq \text{arr.size()} \leq 10^5$

$1 \leq \text{arr}[i] \leq 10^6$

Code:

Output

14. Given an array `arr[]` of N integers and a positive integer K . The task is to cyclically rotate the array clockwise by K .

Note: Keep the first position of array unaltered.

Boundary Condition(s):

$2 \leq N \leq 100$ $-100 \leq arr[i] \leq 100$

$1 \leq K \leq 100$

Input Format:

The first line contains N representing the size of `arr[]`.

The next N lines contain N integers representing the elements of `arr[]`.

The $(N+2)$ th line contains K representing the number of rotations.

Output Format:

The first line contains N integers representing the revised array `arr[]`.

Code:

Output

15. The program must accept an integer matrix of size $R \times C$ as the input. The program must print all possible non-overlapping square submatrices in the given matrix as the output. The non-overlapping square submatrices must be printed in increasing order of size.

Boundary Condition(s):

$2 \leq R, C \leq 30$

Input Format:

The first line contains R and C separated by a space.

The next R lines, each contains C integers separated by a space.

Output Format:

The lines containing the non-overlapping square submatrices.

Code:

Output

16. The program must accept two integer arrays A1 and A2 as the input. The size of the array A1 is M and the size of the array A2 is N. The program must print the longest common subarray between the given two arrays as the output. If two or more subarrays have the same maximum size, then the program must print the first occurring subarray in A1 as the output.

Boundary Condition(s):

$2 \leq M, N \leq 100$

$1 \leq \text{Each integer value} \leq 10^5$

Input Format:

The first line contains M.

The second line contains M integers separated by a space.

The third line contains N.

The fourth line contains N integers separated by a space.

Output Format:

The first line contains the longest common subarray between the given two arrays.

Code:

Output

17. The program must accept two character matrices M1 and M2 as the input. The size of the matrix M1 is $R \times C$ and the size of the matrix M2 is $K \times K$. The program must print M2 if the transpose of M2 is present in M1. Else the program must print -1 as the output.

Boundary Condition(s):

$2 \leq R, C \leq 50$

$2 \leq K \leq \text{Minimum value between } R \text{ and } C$

Input Format:

The first line contains R and C separated by a space.

The next R lines, each contains C characters separated by a space.

The (R+2)th line contains K.

The next K lines, each contains K characters separated by a space.

Output Format:

The first K lines, each contains K characters separated by a space or the first line contains -1.

Code:

Output

18. The program must accept an integer matrix of size $R \times C$ and two integer X, Y as the input. The program must print the sum of integers in all $X \times Y$ non-overlapping submatrices as the output.

Boundary Condition(s):

$2 \leq X \leq R \leq 50$

$2 \leq Y \leq C \leq 50$

$1 \leq \text{Matrix element value} \leq 1000$

Input Format:

The first line contains R and C separated by a space.

The next R lines, each contains C integers separated by a space.

The $(R+2)$ th line contains X and Y separated by a space.

Output Format:

The first line contains the integer values representing the sum of integers in all the $X \times Y$ non-overlapping submatrices.

Code:

Output

19. The program must accept an integer matrix of size $R \times C$ and two integer X, Y as the input. The program must print the sum of integers in all $X \times Y$ non-overlapping submatrices as the output.

Boundary Condition(s):

$2 \leq X \leq R \leq 50$

$2 \leq Y \leq C \leq 50$

$1 \leq \text{Matrix element value} \leq 1000$

Input Format:

The first line contains R and C separated by a space.

The next R lines, each contains C integers separated by a space.

The $(R+2)$ th line contains X and Y separated by a space.

Output Format:

The first line contains the integer values representing the sum of integers in all the $X \times Y$ non-overlapping submatrices.

Code:

Output

20. Mr. Rao is relocating from place A to B. The moving truck has a maximum capacity C. There are 'N' items in the house where each item has a corresponding value (V_i) and weight(W_i). Mr. Rao has to carry only the most valuable items whose total weight does not exceed the capacity of truck. The task here is to find those items (single or combination of items) whose total value (v) will be the maximum and their corresponding weight(w) will not exceed truck capacity(c). Here,

- N= No. of items
- C= Maximum capacity of the truck, an integer value,
- $W[0 \text{ to } N-1]$ - An array consisting weight of each item
- $V[0 \text{ to } N-1]$ – An array consisting value of each item.

Code:

Output